

PCC UNIT-5

1. Discuss the frame structure for GSM.

A. The GSM frame structure is hierarchical, consisting of time slots, frames, multiframe, superframes, and hyperframes. Each level organizes transmission in GSM's TDMA system, ensuring synchronization, efficient resource allocation, and support for signaling and traffic channels.

1. Basic Time Slot

- Fundamental unit of GSM transmission.
- Duration: **577 microseconds (0.577 ms)**.
- Carries **156.25 bits** of information.
- Each RF carrier is divided into **8 time slots**, allowing 8 users to share one frequency channel (TDMA principle).

2. TDMA Frame

- A TDMA frame consists of **8 consecutive time slots**.
- Duration: **4.615 ms**.
- Represents the basic cycle of GSM transmission.

3. Multiframe

- Two types:
 - **Traffic multiframe**: 26 TDMA frames (used for voice/data).
 - Duration: **120 ms**.
 - 24 frames carry traffic, 2 frames used for control/synchronization.
 - **Control multiframe**: 51 TDMA frames (used for signaling/control channels).
 - Duration: **235.4 ms**.
 - Contains broadcast, synchronization, and common control channels.

4. Superframe

- Groups multiframes together.
- **Traffic superframe**: $51 \times 26 = 1326$ TDMA frames.
- Duration: **6.12 seconds**.
- Ensures alignment between traffic and control channels.

5. Hyperframe

- The largest unit in GSM frame hierarchy.
- Contains **2048 superframes**.
- Duration: **3 hours, 28 minutes, 53 seconds**.
- Provides long-term synchronization and is essential for encryption key management.

2. Write about GSM channels?

A. In GSM, **channels** are logical divisions of the radio interface used to carry information. They are broadly classified into **Traffic Channels (TCH)** and **Control**

Channels (CCH).

1. Traffic Channels (TCH)

- Carry **user information** (voice or data).
- Types:
 - **Full-rate TCH (TCH/F)**: One user per time slot.
 - **Half-rate TCH (TCH/H)**: Two users share one time slot.
- Duration: 26-frame multiframe structure.
- Purpose: Provide continuous communication for subscribers.

2. Control Channels (CCH)

Used for signaling, synchronization, and system information. Sub-types:

(a) Broadcast Channels (BCH)

- **Frequency Correction Channel (FCCH)**: Helps mobiles tune to correct frequency.
- **Synchronization Channel (SCH)**: Provides frame synchronization and BS identity.
- **Broadcast Control Channel (BCCH)**: Transmits cell information (cell ID, frequency list, access parameters).

(b) Common Control Channels (CCCH)

- **Paging Channel (PCH)**: Alerts mobiles of incoming calls/messages.
- **Random Access Channel (RACH)**: Used by mobiles to request access (uplink).
- **Access Grant Channel (AGCH)**: Grants access to mobiles, assigning dedicated channels.

(c) Dedicated Control Channels (DCCH)

- **Standalone Dedicated Control Channel (SDCCH)**: Used for signaling before a call is set up (authentication, location update, SMS).
- **Slow Associated Control Channel (SACCH)**: Carries continuous control info (power control, timing advance).
- **Fast Associated Control Channel (FACCH)**: Used for urgent signaling (handoff); temporarily replaces traffic channel.

3. Compare LTE and WiMAX.

A.

Aspect	LTE (Long Term Evolution)	WiMAX (Worldwide Interoperability for Microwave Access)
Standard	3GPP (3rd Generation Partnership Project)	IEEE 802.16e (Mobile WiMAX)
Channel Bandwidth	Flexible: 1.4 MHz to 20 MHz	Fixed: 5, 10, 20 MHz
Access Technique	OFDMA (downlink), SC-FDMA (uplink)	OFDMA (both uplink and downlink)
Peak Data Rate	~100 Mbps (DL), ~50 Mbps (UL) in Release 8; higher with LTE-Advanced	~70 Mbps (DL) with 20 MHz channel
Latency	~10 ms or less	~30–50 ms
Mobility Support	Strong support for handoff, roaming, and high-speed mobility	Limited mobility; better suited for fixed/nomadic broadband
Architecture	All-IP, evolved packet core (EPC), supports VoLTE	IP-based, ASN + CSN architecture
Deployment	Widely adopted globally by telecom operators	Limited adoption; used in niche broadband deployments
Spectrum Efficiency	Higher, due to flexible bandwidth and advanced MIMO	Moderate, less flexible spectrum use
Voice Support	VoLTE (Voice over LTE) with HD quality	VoIP over WiMAX (less standardized)
Evolution	LTE-Advanced, LTE-Advanced Pro, foundation for 5G	WiMAX lost momentum; replaced by LTE in most markets

4. How does digital cellular technology improve call quality and data transfer rates compared to analog systems?

A. 1. Call Quality Improvements

- **Noise Immunity:**
 - Analog signals degrade with noise and interference.
 - Digital signals use coding and error correction, making them more resistant to distortion.
- **Compression & Coding:**
 - Voice is digitized and compressed, reducing redundancy.
 - Error detection/correction ensures clearer audio.
- **Consistent Quality:**
 - In analog, signal fades gradually with distance.
 - In digital, quality remains stable until the signal drops below a threshold.
- **Encryption:**
 - Digital systems allow secure voice transmission, preventing

eavesdropping.

2. Data Transfer Rate Improvements

- **Efficient Spectrum Use:**
 - Digital systems (TDMA, CDMA, OFDMA) allow multiple users to share the same frequency band efficiently.
 - This increases capacity and throughput.
- **Support for Non-Voice Services:**
 - Analog systems were voice-only.
 - Digital systems support SMS, MMS, internet browsing, and broadband data.
- **Higher Bandwidth Utilization:**
 - Techniques like CDMA and OFDMA spread signals across wide bandwidths, enabling higher data rates.
- **Adaptive Modulation:**
 - Digital systems adjust modulation schemes (e.g., QPSK, QAM) based on channel conditions to maximize speed.

3. Network Performance Enhancements

- **Handoff Efficiency:**
 - Digital systems support soft handoff (CDMA) and seamless mobility.
- **Capacity Expansion:**
 - More users per cell compared to analog.
- **Integration:**
 - Digital networks integrate voice and data, paving the way for 3G, 4G, and 5G services.

5. Explain the principle of WCDMA and write its advantages and disadvantages.

A. Principle of WCDMA (Wideband Code Division Multiple Access)

- **Definition:** WCDMA is a **3G mobile communication technology** based on CDMA but with a wider bandwidth (5 MHz per carrier). It is the core air interface of **UMTS (Universal Mobile Telecommunications System)**.
- **Principle of Operation:**
 1. **Spread Spectrum:**
 - Each user's data is spread over a wide bandwidth using unique codes.
 - Signals overlap in frequency and time but are distinguished by correlation with their codes.
 2. **Wideband Channel:**
 - Uses 5 MHz channels, allowing high data rates compared to narrowband CDMA.
 3. **Soft Capacity:**
 - Capacity depends on interference levels, not fixed slots.
 - More users increase noise, gradually reducing quality.
 4. **Soft Handoff:**
 - A mobile can connect to multiple base stations simultaneously,

ensuring seamless mobility.

5. Power Control:

- Essential to solve the near–far problem by equalizing received signal strengths.

Advantages of WCDMA

1. High Data Rates:

- Supports up to 2 Mbps (initial 3G) and higher with HSPA evolution.
- Enables video calls, mobile internet, and multimedia services.

2. Efficient Spectrum Use:

- Multiple users share the same frequency band using codes.

3. Improved Voice Quality:

- Wideband spreading reduces interference and fading.

4. Soft Handoff:

- Reduces dropped calls during mobility.

5. Global Standard:

- Adopted widely as part of UMTS, ensuring interoperability.

6. Flexibility:

- Supports both circuit-switched (voice) and packet-switched (data) services.

Disadvantages of WCDMA

1. Complexity:

- Requires advanced receivers, correlation, and synchronization.

2. Interference-Limited:

- Capacity decreases as more users join due to rising noise.

3. Near–Far Problem:

- Strong signals can overpower weak ones unless strict power control is maintained.

4. High Deployment Cost:

- Needs new infrastructure compared to GSM/TDMA systems.

5. Battery Drain:

- Power control and wideband processing consume more energy in mobile devices.

6. List the GSM specifications.

A. GSM Specifications

1. Frequency Bands

- GSM 900: Uplink 890–915 MHz, Downlink 935–960 MHz
- GSM 1800 (DCS): Uplink 1710–1785 MHz, Downlink 1805–1880 MHz
- GSM 1900 (PCS): Uplink 1850–1910 MHz, Downlink 1930–1990 MHz

2. Channel Bandwidth

- Each carrier: **200 kHz** wide
- Divided into **8 time slots** (TDMA principle)

3. Modulation

- **GMSK (Gaussian Minimum Shift Keying)**

- Provides spectral efficiency and constant envelope for power-efficient transmission
- 4. Multiple Access**
 - **TDMA (Time Division Multiple Access)**
 - 8 users per carrier frequency
- 5. Frame Structure**
 - Time slot: 577 μ s
 - TDMA frame: 4.615 ms (8 slots)
 - 26-frame multiframe (traffic) = 120 ms
 - 51-frame multiframe (control) = 235.4 ms
 - Superframe = 6.12 s
 - Hyperframe = 3 h 28 min 53 s
- 6. Data Rates**
 - Full-rate speech channel: 13 kbps
 - Half-rate speech channel: 6.5 kbps
 - Control channel: 22.8 kbps gross bit rate
- 7. Duplexing**
 - **FDD (Frequency Division Duplex)**
 - Separate uplink and downlink bands with 45 MHz spacing (in GSM 900)
- 8. Cell Structure**
 - Hexagonal cells with frequency reuse
 - Supports handoff between cells
- 9. Services**
 - Voice (circuit-switched)
 - SMS (Short Message Service)
 - Data (up to 9.6 kbps in early GSM, later enhanced with GPRS/EDGE)

7. What are the key features and benefits of LTE compared to previous wireless technologies?

A. Key Features of LTE

- 1. High Data Rates**
 - Downlink: up to **100 Mbps** (Release 8), higher with LTE-Advanced.
 - Uplink: up to **50 Mbps**.
 - Far superior to 3G WCDMA (~2 Mbps) and GSM (~9.6 kbps).
- 2. Low Latency**
 - End-to-end latency reduced to **<10 ms**, compared to 50–100 ms in 3G.
 - Critical for real-time applications like video calls, gaming, and IoT.
- 3. All-IP Network Architecture**
 - Voice, data, and multimedia carried over IP.
 - Simplifies integration with internet services and supports VoLTE (Voice over LTE).
- 4. Flexible Spectrum Use**
 - Operates in bandwidths from **1.4 MHz to 20 MHz**.
 - Supports both FDD (Frequency Division Duplex) and TDD (Time Division

Duplex).

5. Advanced Modulation & Multiplexing

- Downlink: **OFDMA (Orthogonal Frequency Division Multiple Access)**.
- Uplink: **SC-FDMA (Single Carrier FDMA)** for power efficiency.
- Adaptive modulation (QPSK, 16QAM, 64QAM) based on channel conditions.

6. Improved Spectral Efficiency

- More bits transmitted per Hz compared to GSM, CDMA, and WCDMA.
- Supports higher user density per cell.

7. MIMO (Multiple Input Multiple Output)

- Uses multiple antennas for spatial multiplexing.
- Enhances throughput and reliability.

Benefits Compared to Previous Technologies

- **Higher Capacity:** Supports more users and higher traffic volumes.
- **Better Call Quality:** VoLTE provides HD voice with lower delay.
- **Broadband Data Services:** Enables streaming, video conferencing, and mobile internet at speeds comparable to wired broadband.
- **Seamless Mobility:** Smooth handoffs and global roaming support.
- **Future-Proof:** Designed to evolve into LTE-Advanced and 5G, ensuring long-term scalability.
- **Energy Efficiency:** Reduced transmission time and efficient uplink design extend battery life.

8. Describe the features and services GSM.

A. Features of GSM

1. Digital System

- GSM is a fully digital cellular system (2G).
- Uses **TDMA (Time Division Multiple Access)** with 8 time slots per carrier.

2. Frequency Bands

- GSM 900, GSM 1800, GSM 1900 (regional variations).
- Each carrier has **200 kHz bandwidth**.

3. Modulation

- **GMSK (Gaussian Minimum Shift Keying)** for efficient spectrum use and constant envelope transmission.

4. International Standardization

- GSM is a global standard, ensuring interoperability and roaming across countries.

5. Security

- Subscriber identity protected by **SIM card**.
- Authentication and encryption mechanisms for secure communication.

6. Frame Structure

- Hierarchical: time slot → frame → multiframe → superframe → hyperframe.

- Ensures synchronization and efficient multiplexing.

7. Duplexing

- **FDD (Frequency Division Duplex)** with separate uplink and downlink bands.

8. Capacity

- Supports multiple users per frequency channel (8 per carrier).
- Frequency reuse across cells increases overall capacity.

Services of GSM

1. Voice Services

- High-quality digital voice transmission.
- Full-rate and half-rate speech channels.

2. Data Services

- Circuit-switched data up to **9.6 kbps** (basic GSM).
- Later enhanced with **GPRS** and **EDGE** for higher speeds.

3. Messaging Services

- **SMS (Short Message Service)**: Up to 160 characters.
- **Cell Broadcast Service (CBS)**: Broadcast messages to multiple users in a cell.

4. Supplementary Services

- Call forwarding, call waiting, call barring, caller ID.
- Conference calling and multiparty services.

5. Roaming Services

- International roaming supported due to global standardization.
- SIM card portability across networks.

6. Emergency Services

- Access to emergency numbers (e.g., 112/911) even without SIM.

9. Discuss about architecture of WiMAX in detail.

A. WiMAX (Worldwide Interoperability for Microwave Access) architecture is an IP-based framework designed to deliver broadband wireless access over large areas. It is structured into three main components: Mobile Station (MS), Access Service Network (ASN), and Connectivity Service Network (CSN). Together, they ensure end-to-end communication, mobility, and interoperability.

1. Mobile Station (MS)

- **Definition:** The subscriber device (laptop, smartphone, CPE) that connects to the WiMAX network.
- **Functions:**
 - Provides wireless access to end users.
 - Implements the physical (PHY) and MAC layers of IEEE 802.16.
 - Handles authentication and encryption.
 - Supports mobility features in Mobile WiMAX (802.16e).

2. Access Service Network (ASN)

- **Definition:** The radio access part of WiMAX, consisting of **Base Stations (BS)** and **ASN Gateways (ASN-GW)**.

- **Components & Functions:**

- **Base Station (BS):**

- Provides wireless coverage.
 - Implements radio resource management, scheduling, and QoS.
 - Communicates with MS using OFDMA (Orthogonal Frequency Division Multiple Access).

- **ASN Gateway (ASN-GW):**

- Manages mobility (handoffs between BSs).
 - Provides security and AAA (Authentication, Authorization, Accounting).
 - Acts as a bridge between the ASN and CSN.

- **Role:** Ensures seamless connectivity and mobility within the access network.

3. Connectivity Service Network (CSN)

- **Definition:** The IP-based core network of WiMAX.

- **Functions:**

- Provides internet connectivity and access to external networks (PSTN, VoIP, corporate VPNs).
 - Handles subscriber management, billing, and roaming.
 - Supports IP services such as DHCP, DNS, and AAA servers.
 - Enables interworking with other networks (e.g., GSM/UMTS).

4. Protocol Layers

- **Physical Layer (PHY):**

- Uses OFDM/OFDMA for robust transmission.
 - Supports both Line-of-Sight (LOS) and Non-LOS communication.

- **MAC Layer:**

- Divided into sublayers:
 - **Convergence Sublayer (CS):** Maps higher-layer protocols (IP, Ethernet) into MAC.
 - **Common Part Sublayer (CPS):** Handles scheduling, bandwidth allocation, and QoS.
 - **Security Sublayer:** Provides authentication and encryption.

5. Key Features of WiMAX Architecture

- **IP-based design:** Supports voice, video, and data over a unified framework.
- **QoS Support:** Ensures service differentiation for real-time and non-real-time applications.
- **Mobility:** Mobile WiMAX (802.16e) supports handoffs and roaming.
- **Scalability:** Covers up to 50 km range with speeds up to 70 Mbps.
- **Interoperability:** Standardized by WiMAX Forum to ensure compatibility across vendors.

6. Advantages

- Wide coverage compared to Wi-Fi.
- High data rates suitable for broadband access.
- Supports both fixed and mobile deployments.
- Cost-effective alternative to wired broadband in rural areas.

10. Explain the architecture of GSM.

A. GSM Architecture

The **Global System for Mobile Communication (GSM)** architecture is divided into several subsystems that work together to provide mobile services. The main components are:

1. Mobile Station (MS)

- **Definition:** The subscriber's handset + SIM card.
- **Functions:**
 - Provides user interface (voice, SMS, data).
 - SIM card stores subscriber identity (IMSI), authentication keys, and service information.
 - Handles radio transmission with the Base Station Subsystem.

2. Base Station Subsystem (BSS)

- **Definition:** Manages radio communication between MS and the network.
- **Components:**
 - **Base Transceiver Station (BTS):**
 - Handles radio transmission/reception with mobiles.
 - Covers one cell area.
 - **Base Station Controller (BSC):**
 - Controls multiple BTSs.
 - Manages radio resources, handovers, and frequency allocation.
- **Role:** Provides the air interface and manages radio resources.

3. Network Switching Subsystem (NSS)

- **Definition:** The core of GSM, responsible for call switching and subscriber management.
- **Components:**
 - **Mobile Switching Center (MSC):** Central switching node; manages calls, SMS, and mobility.
 - **Home Location Register (HLR):** Database of permanent subscriber information (IMSI, services, location).
 - **Visitor Location Register (VLR):** Temporary database for subscribers currently in the MSC area.
 - **Authentication Center (AuC):** Provides authentication and encryption keys.
 - **Equipment Identity Register (EIR):** Stores IMEI numbers; blocks stolen/unauthorized devices.
- **Role:** Ensures call routing, mobility management, and subscriber authentication.

4. Operation and Support Subsystem (OSS)

- **Definition:** Provides tools for network operators to monitor, control, and maintain the GSM system.
- **Functions:**
 - Network management, fault detection, performance monitoring.

- Ensures smooth operation and optimization of the GSM network.

5. Interfaces

- **Air Interface (Um):** Between MS and BTS.
- **Abis Interface:** Between BTS and BSC.
- **A Interface:** Between BSC and MSC.
- Other interfaces connect NSS databases (HLR, VLR, AuC, EIR).

11. What are the key differences between 2G, 3G, 4G, and 5G digital cellular systems?

Generation	Technology Standard	Data Rate	Access Technique	Latency	Services	Key Features
2G	GSM, CDMA (digital)	~9.6 kbps (basic), up to 384 kbps with GPRS/EDGE	TDMA, FDMA	~300–500 ms	Voice, SMS, basic data	Digital voice, SMS, SIM authentication
3G	UMTS (WCDMA), CDMA2000	Up to 2 Mbps (basic), ~14 Mbps with HSPA	CDMA (wideband)	~100–200 ms	Voice, video calls, mobile internet	Multimedia support, packet-switched data
4G	LTE, WiMAX	100 Mbps (mobile), up to 1 Gbps (stationary)	OFDMA (DL), SC-FDMA (UL)	~30–50 ms	HD voice (VoLTE), broadband internet, video streaming	All-IP network, high spectral efficiency, MIMO
5G	NR (New Radio)	1–10 Gbps peak	OFDMA + Massive MIMO + Beamforming	~1 ms	Enhanced broadband, IoT, AR/VR, autonomous systems	Ultra-low latency, network slicing, massive device connectivity

12. Discuss 5G Technology in detail and mention its advantages and disadvantages.

A. 5G is the fifth generation of mobile communication technology, designed to deliver ultra-fast data speeds, extremely low latency, massive device connectivity, and support for advanced applications like IoT, autonomous vehicles, and smart cities. Compared to 4G LTE, it represents a major leap in performance and

capabilities.

Features of 5G Technology

1. High Data Rates

- Peak download speeds up to **10 Gbps** (much higher than 4G LTE's ~100 Mbps).
- Enables seamless streaming of 4K/8K video, AR/VR, and cloud gaming.

2. Ultra-Low Latency

- Latency reduced to **1 ms or less**, critical for real-time applications like autonomous driving, remote surgery, and industrial automation.

3. Massive Connectivity

- Supports up to **1 million devices per square kilometer**, ideal for IoT ecosystems and smart cities.

4. Enhanced Spectrum Use

- Operates in **sub-6 GHz bands** and **millimeter wave (24–100 GHz)** for high capacity.
- Uses advanced techniques like **beamforming** and **massive MIMO**.

5. Network Slicing

- Allows operators to create multiple virtual networks on the same physical infrastructure, tailored for specific services (e.g., healthcare, manufacturing, entertainment).

6. Energy and Spectrum Efficiency

- More efficient than 4G, reducing energy consumption per bit transmitted.
- Improves overall network performance.

Advantages of 5G

- **Extremely High Speed:** Enables broadband-like mobile internet.
- **Low Latency:** Supports mission-critical applications (autonomous vehicles, remote surgery).
- **Massive IoT Support:** Connects billions of devices simultaneously.
- **Improved Reliability:** Advanced error correction and robust connectivity.
- **Enhanced User Experience:** Smooth video streaming, AR/VR, and gaming.
- **Economic Growth:** Drives innovation in industries like healthcare, transport, manufacturing, and education.

Disadvantages of 5G

- **Infrastructure Cost:** Requires new base stations, fiber backhaul, and dense small-cell deployment.
- **Limited Coverage Initially:** Millimeter waves have short range and poor penetration (walls, buildings).
- **Device Compatibility:** Needs new 5G-enabled smartphones and IoT devices.
- **Energy Consumption:** Although efficient per bit, dense networks may increase overall power usage.
- **Security Concerns:** Larger attack surface due to massive IoT connectivity.
- **Health & Environmental Debates:** Public concerns about radiation exposure, though scientific consensus finds no proven harm.